# MARKOV_FIBRE_FREQ

# MARKOV_FIBRE_FREQ REPORT

# RESULTS: MARKOV_FIBRE_FREQ

PREREG.md sha256 a56c5e85e7cd004abc56e40b0a5a86783b383f150ba5b90ec122832237200cc5. It was hashed after exploration (data/explore.sha256) and before the holdout run (scripts/test.py → data/test.json, logs/test.log).

Holdout: 3000 seeds ‘mfholdout-d’ × 539 steps = 1,617,000 steps, giving 179,050 hits (hit rate 0.11073).

| fibre β | predicted (×1/32) | predicted | measured | count |
|---|---|---|---|---|
| 0 | 1 | 0.03125 | 0.03205 | 5739 |
| 1 | 8 | 0.25000 | 0.24959 | 44689 |
| 2 | 3 | 0.09375 | 0.09469 | 16955 |
| 3 | 0 | 0 | 0.00012 | 21 (prefix) |
| 4 | 7 | 0.21875 | 0.21654 | 38772 |
| 5 | 8 | 0.25000 | 0.25008 | 44777 |
| 6 | 5 | 0.15625 | 0.15677 | 28069 |
| 7 | 0 | 0 | 0.00016 | 28 (prefix) |

| Test | Measured | Threshold | Verdict |
|---|---|---|---|
| M1: $\chi^2$ on the 6 open fibres vs prediction | 9.80 | ≤ 20.52 | PASS |
| M2: share of fibres {3,7} | $2.7 \cdot 10^{-4}$ | $\leq 10^{-3}$ | PASS |
| M3: uniform-on-six null rejected (non-vacuity) | $\chi^2 = 42\,684$ | > 20.52 | PASS |

**Overall: PASS (Category B, numerical).** The result is non-vacuous: the relative rates on {0,1,2,4,5,6} = 1:8:3:7:8:5 are predicted and confirmed. **Bridge: NO.** This is residual-internal only.

Post-test exact check (scripts/exact.py, logs/exact.log; sympy rational solve at M = 216): fibre law = (1/32, 1/4, 3/32, 0, 7/32, 1/4, 5/32, 0) and hit rate = 1/9 exactly. Both are Category A as statements about the lifted chain. # NOTES: MARKOV_FIBRE_FREQ ## Lemmas - **L1 (Category A, elementary).** On a hit u = 21 + 9j, j ≡ u − 5 (mod 8). So β̂ is exactly the class of u mod 8, and the fibre law among hits is a 2-adic question. - **L2 (Category A, about the lifted chain; exact sympy solve, logs/exact.log).** - The lifted chain mod 216 has a unique closed class {u ≢ 0 mod 4}. This is Lemma V. - Its stationary law factorizes: (u mod 8) is independent of (u mod 27). The mod-9 marginal is uniform, giving hit rate 1/9. - The u mod 8 law is (0, 7, 8, 5, 0, 1, 8, 3)/32 for u ≡ 0..7. - Hence the fibre law among hits is (1, 8, 3, 0, 7, 8, 5, 0)/32. - Refining to 2^k·27 for k = 5, 7, 9, 11 leaves it unchanged. The truncation is already exact at 2^3. - **L3 (Category B).** Real T3 orbits from SHA3 seeds follow this law (holdout $\chi^2 = 9.8$, df = 5). The open assumption is the heuristic that the branch (u mod 3) is equidistributed and independent of the low 2-adic digits. That is the uniform-lift step, and it is not proved for actual orbits. - The zeros at fibres 3 and 7 are Lemma V. The nontrivial part is the 1:8:3:7:8:5 pattern, coming from the 2-adic Markov chain on u mod 8 driven by a uniform branch.

# Status

- Path → fibre is now predictable from first principles (residual-internal).
- Still OPEN: C6, C10, C11 and E1–E3. They are blocked on Arvin accepting a Π constraint / grading and a T3 q-series. No Residual ↔ Resonant map was tested, and no Resonant objects were created.

# Next step

Prove L3 conditionally, or test it directly. Measure the joint law of (u mod 3, u mod 8) on orbits to check the independence that the uniform-lift step assumes. Then, under the lock, hash-preregister a map that compares the (1,8,3,0,7,8,5,0)/32 law against a pre-specified Resonant-side fibre weighting. That is the first object that could be a non-vacuous Residual ↔ Resonant test, and it needs Arvin's sign-off.

CORE_FREEZE respected. Only this new folder was written. No push and no send. Order28, RTA_FOUNDATION_BOUND and the order* folders were not touched.